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LH The Life and Death of Stars

Lecture 4

(JCD 3)

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Previous lecture

- Astrometry determines position and distance
- Photometry determines brightness, luminosity, magnitude, colour
- Observational HR diagram through colour-magnitude diagram
- Spectroscopy determines effective temperature, gravity, elemental abundances, rotation rate
- Masses, radii and ages usually determined via stellar modeling
- Asteroseismology for most precise masses and ages

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Outline

Equation of state

Ideal gas

Radiation pressure

Degenerate matter

Summary

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Equations of state

- Thermodynamic properties of gas
- Assumption that the gas in a star is in thermodynamic equilibrium at any point
- Relates local state variables: temperature T, pressure P, density ρ, internal energy density u, chemical composition $\Rightarrow$ Local Thermodynamic Equilibrium - LTE
- Stellar interiors often simplified by assuming fully ionised gas and no particle interactions $\rightarrow$ Ideal gas

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Recap: internal energy density and pressure

- Need to distinguish between relativistic and non-relativistic motion
- Non-relativistic:

$$u = \frac{3}{2}P$$

- Relativistic:

$$u = 3P$$

Simple gas

- Assume ideal gas with single type non-relativistic particles
- Ideal gas law links pressure P, temperature T, and volume V:

$$P\,V = N\,k\,T$$

  - N: number of particles
  - k the Boltzmann constant
- In astronomy, we use density rather than volume.

$$V = \frac{M}{\rho} = \frac{N\overline{m}}{\rho}$$

  - $\overline{m}$: average mass of a gas particle
- Thus:

$$P = \frac{\rho\,k\,T}{\overline{m}}$$

Mean molecular weight

- We define the mean molecular weight as

$$\mu \equiv \frac{\overline{m}}{m_u}$$

- Equation state of gas:

$$\boxed{P = \frac{\rho\, k\, T}{\mu\, m_u}}$$

- m_u = atomic mass unit (amu) $= 1.66 \cdot 10^{-27}$ kg
- In other words: mean molecular weight is the average mass of a free particle in the gas, expressed in atomic mass units
- For a simple (one atomic) gas, μ equals the atomic mass

Gas mixture

- Generally, gases are atomic (neutral, partially ionised, or fully ionised) except for molecules that are present in cool star atmospheres
  - Neutral gas: free particles are the atomic nuclei
  - Fully ionised gas: free particles are the atomic nuclei and all the free electrons
  - Remember: $m_p \approx m_n \gg m_e$
- Examples:
  - Neutral hydrogen: $\mu = 1$
  - Neutral helium: $\mu = ...$
  - Ionised hydrogen: $\mu = 1/2$ (total mass $\sim$ 1amu, 2 particles ($1H^+ + 1e^-$))
  - Fully ionised helium: $\mu = ...$
  - Fully ionised heavy element: $\mu \approx ...$

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  - Remember: $m_p \approx m_n \gg m_e$
- Examples:
  - Neutral hydrogen: $\mu = 1$
  - Neutral helium: $\mu = 4$
  - Ionised hydrogen: $\mu = 1/2$ (total mass $\sim$ 1amu, 2 particles ($1\mathrm{H}^+ + 1\mathrm{e}^-$))
  - Fully ionised helium: $\mu = 4/3$
  - Fully ionised heavy element: $\mu \approx 2$ (for heavy elements: $\#p \approx \#n \approx \#e$)

Fully ionised ideal gas

- Same gas law - more complicated mean molecular weight:

$$\mu^{-1} = \sum_j X_j \frac{Z_j + 1}{A_j}$$

  with Z_j the atomic number, A_j the atomic mass and X_j the mass fraction of an element respectively.
- Remember, we define mass fractions of different elements:
  - X =(total mass of hydrogen)/(total mass of star)
  - Y =(total mass of helium)/(total mass of star)
  - Z =(total mass of metals)/(total mass of star)
  - $X + Y + Z = 1$
- You can show that

$$\mu \approx \frac{4}{3 + 5X - Z}$$

Examples

- For the Sun: $X = 0.73, Y = 0.25, Z = 0.02$
  - Solar core (ionised): $\mu \approx 0.6$
  - Solar atmosphere (neutral): $\mu \approx 1.2$ [verify!]

Electron gas

- Mean molecular weight per electron defined through electron number density n_e:
$$n_e = \frac{\rho}{\mu_e m_u}$$
- You can show that
$$\mu_e = \frac{2}{1+X}$$
- Important for eg electron degenerate matter

Recap on gas energetics

- First law of thermodynamics: $dQ = dU + PdV$
- If volume does not change $\rightarrow dQ = c_V\, dT$ with c_V the specific heat at constant volume.
  - Verify that for ideal gas $c_V = \frac{3k}{2\mu m_u}$
- If pressure does not change $\rightarrow dQ = c_P\, dT$ with c_P the specific heat at constant pressure.
  - Verify thatfor ideal gas $c_P = \frac{5k}{2\mu m_u}$
- Define $\gamma = c_P/c_V$. For an ideal gas $\gamma = 5/3$.
- Adiabatic process if there is no heat exchange. Thus $dQ = 0$:

$$c_V\, dT = -P\, dV$$

$$\Rightarrow \frac{dP}{P} = -\gamma \frac{dV}{V} = \gamma \frac{d\rho}{\rho}$$

Adiabatic exponents

- Three interdependent quantities relating pressure, density and temperature under general circumstances

$$\Gamma_1 = \left(\frac{\partial \ln P}{\partial \ln \rho}\right)_s$$

$$\frac{\Gamma_2}{\Gamma_2 - 1} = \left(\frac{\partial \ln P}{\partial \ln T}\right)_s$$

$$\Gamma_3 - 1 = \left(\frac{\partial \ln T}{\partial \ln \rho}\right)_s$$

- Verify that

$$\frac{\Gamma_2}{\Gamma_2 - 1} = \frac{\Gamma_1}{\Gamma_3 - 1}$$

- For an ideal gas, it holds that $\Gamma_1 = \Gamma_2 = \Gamma_3 = \gamma = 5/3$

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Radiation pressure

In addition to gas pressure (from ions and electrons), there is also radiation pressure (from photons)

$$P_{\text{rad}} = \frac{1}{3} a\, T^4$$

where $a = 4\sigma/c = 7.56 \cdot 10^{-16}\,\mathrm{N\,m^{-2}\,K^{-4}}$ is the radiation constant.

- Photons move relativistically, thus $u_{\text{rad}} = a\, T^4$
- Total pressure is sum of gas pressure and radiation pressure (same for internal energy density)
- Examples:
  - The Sun ($T \approx 6000\,\mathrm{K}$): $P_{\text{rad}} \approx 0.3\,\mathrm{N\,m^{-2}}$
  - O star ($T \approx 40000\,\mathrm{K}$): $P_{\text{rad}} \approx 650\,\mathrm{N\,m^{-2}}$

Adiabatic exponent for photons

- Photons move relativistically, thus $U_{\text{rad}} = 3P\,V$
- In adiabatic process for extremely relativistic particles:

$$\frac{dP}{P} = \frac{4}{3}\frac{d\rho}{\rho}$$

$$\Rightarrow \Gamma_1 = \frac{4}{3}$$

- It can be shown that, for radiation pressure:

$$\Gamma_1 = \Gamma_2 = \Gamma_3 = \frac{4}{3}$$

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Degenerate matter

- In low temperature, high density limit, need for quantum-mechanical effects
- Use Fermi-Dirac statistics to describe number density of particles.
- Complete degeneracy for $T \to 0$
- Fermi momentum and Fermi energy such that for particles with $p < p_F$

$$n = \frac{8\pi p^3}{3h^3}$$

  and $n = 0$ otherwise.
- Need to distinguish between non-relativistic and relativistic degeneracy pressure
  - Non-relativistic:

$$P_{\rm deg} = \frac{2}{3} u_{\rm deg} \propto \rho^{5/3}$$

  - Relativistic:

$$P_{\rm deg} = \frac{1}{3} u_{\rm deg} \propto \rho^{4/3}$$

Summary

- Ideal gas law:

$$P = \frac{\rho\, k\, T}{\mu\, m_u}$$

- Mean molecular weight:

$$\mu^{-1} = \sum_j X_j \frac{Z_j + 1}{A_j}$$

- Radiation pressure:

$$P_{\mathrm{rad}} = \frac{1}{3} a\, T^4$$

- Degenerate matter: $P_{\mathrm{deg,nr}} \propto \rho^{5/3}$ and $P_{\mathrm{deg,r}} \propto \rho^{4/3}$

NEXT LECTURE: Hydrostatic equilibrium, mass equation, estimates for internal pressure and temperature